Institute of Engineering and Technology (IET)

Customer Churn Prediction & ROI-Backed Retention Playbook

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ABSTRACT

Customer churn is a major challenge for subscription-based businesses, especially in the telecom sector where retaining customers is significantly more cost-effective than acquiring new ones. This project develops a complete machine learning-driven churn prediction and analysis system using the Telco Customer Churn dataset from Kaggle. After data preprocessing, feature engineering, and experimentation with Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, **the XGBoost model emerged as the best performer**, achieving the highest **accuracy (73.7%) and F1-score (0.750)** while maintaining strong recall, making it effective at identifying churn-prone customers.

Model explainability was incorporated using **SHAP**, revealing key churn drivers such as **contract type, tenure, monthly charges, payment method, and technical support usage**. The entire pipeline was deployed as an interactive **Streamlit application**, supporting both single and batch predictions, auto-generated SHAP visualizations, and a dashboard for churn insights. A fully functional **ROI Calculator** was also developed, enabling businesses to estimate the monetary impact of retention campaigns based on churn probability, customer lifetime value (CLV), and intervention cost.

Overall, the final system operates as an **end-to-end churn intelligence and decision-support platform**, combining prediction, interpretability, visualization, and financial analysis to help organizations design data-driven retention strategies.

1.INTRODUCTION

1.1 Background and Motivation

Customer churn is one of the most critical challenges faced by subscription-driven businesses such as telecom, SaaS, and streaming services. Retaining existing customers is significantly more cost-effective than acquiring new ones, with studies showing that acquisition costs can be **5 to 10 times higher**. Moreover, even a small improvement in customer retention can increase profits by **25–95%**.
To address this challenge, identifying customers who are likely to discontinue their services is essential. Machine learning–based churn prediction provides a proactive approach, enabling organizations to intervene early and take data-driven decisions to enhance customer lifetime value.

1.2 Problem Statement

Although companies collect substantial customer data, many lack the capability to **accurately predict churn** or understand **why** customers leave. Traditional analytics techniques often fail to capture complex patterns, lack interpretability, and do not provide financial insights needed for strategic retention planning.
This project addresses these gaps by developing a complete churn prediction and analysis system that combines **machine learning, explainability (SHAP), dashboards, batch prediction capabilities, and an ROI calculator** to support operational and strategic decisions.

1.3 Objectives

• Develop and evaluate ML models (Logistic Regression, Random Forest, Gradient Boosting, XGBoost) to predict customer churn.
• Identify major churn drivers using SHAP explainability and feature analysis.
• Compare models using key metrics and finalize the best-performing model (XGBoost).
• Deploy the complete prediction pipeline in a Streamlit application for single and batch predictions.
• Generate actionable insights through dashboards and SHAP visualizations.
• Build an ROI calculator to measure the financial impact of retention strategies..

1.4 Scope of Work

• Data preprocessing, EDA, and feature engineering
• Model development, tuning, and evaluation
• SHAP-based model explainability
• Streamlit deployment with single/batch prediction
• Dashboards and SHAP visual reports
• ROI calculator integrated into the app

2. Literature Review

Research paper 1:

Customer Churn Prediction in Telecom Using Machine Learning on a Big Data Platform
researchpaper1

Key Algorithms Used

This study evaluated four tree-based machine learning algorithms for predicting customer churn in the telecom sector:

- Decision Tree
- Random Forest
- Gradient Boosted Machine (GBM)
- Extreme Gradient Boosting (XGBoost)

Among these, XGBoost demonstrated the highest predictive performance and was chosen for deployment.

Accuracy Achieved

The model achieved strong predictive accuracy, evaluated primarily by the Area Under the Receiver Operating Characteristic Curve (AUC):

Algorithm	AUC on Main Dataset	AUC on Unseen/Test Dataset
XGBoost	93.3%	89%
Gradient Boosted Machine (GBM)	90.9%	Not reported
Random Forest	87.8%	Not reported
Decision Tree	83%	Not reported

The addition of Social Network Analysis (SNA) features, derived from customers' social connections, significantly improved AUC scores from 84% up to 93.3%.

Business Impact

The predictive model helps telecom operators proactively identify customers likely to churn, enabling targeted retention campaigns. Retaining customers is crucial because:

- Acquiring a new customer costs approximately six times more than retaining an existing one.
- The model's successful early churn prediction enabled SyriaTel to reduce customer churn by about 1.5%, resulting in increased revenue and more efficient marketing spend.

- Integration of big data tools allowed handling over 70 TB of diverse customer data collected over nine months, enhancing feature richness and model accuracy.

This research demonstrates that combining machine learning algorithms with big data platforms and social network analysis offers a powerful solution to reduce churn in telecom companies, improving profitability and customer loyalty.

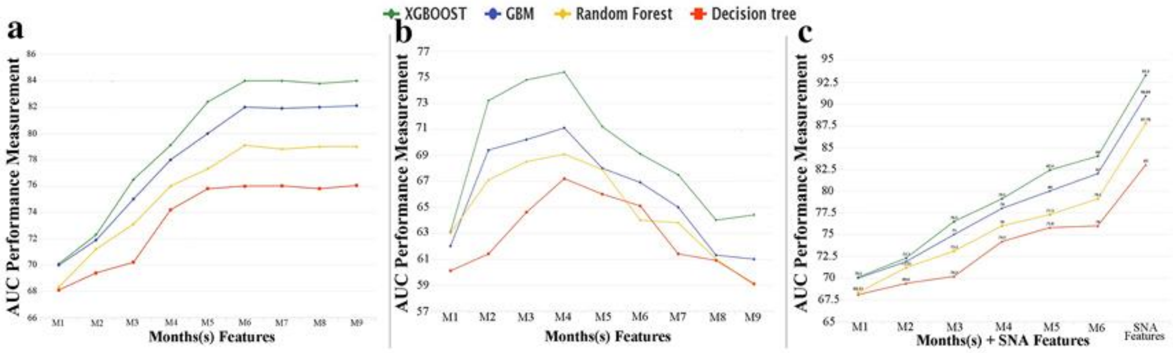

Fig.1: Performance of classification algorithms per sliding window and feature type. Panel (a) shows the improvement of churn predictive model using Statistical Features related to different historical periods, panel (b) presents the changes in predictive model improvement using SNA Features related to the same historical periods, and panel (c) presents the enhancement of churn predictive model when using both statistical and SNA Features

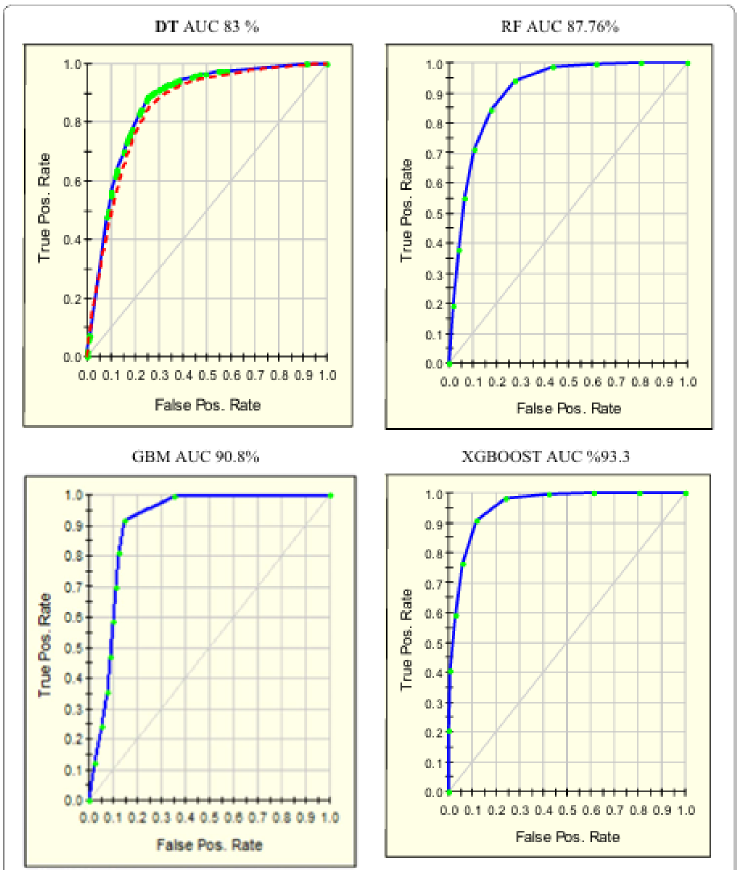

Fig. 12 Receiver operating characteristic curve for each classification algorithm

Research paper 2:

Review on Machine Learning Methods for Customer Churn Prediction

[Researchpaper2]

Key Algorithms Discussed

The paper explores a broad range of machine learning techniques commonly used in churn prediction, including:

- Logistic Regression
- Decision Trees and Ensemble Methods (Random Forest, Gradient Boosting, XGBoost)
- Support Vector Machines (SVM)
- Deep Learning Models (ANN, CNN, RNN)

Among them, ensemble tree-based models such as XGBoost and gradient boosting methods consistently achieve superior accuracy. Deep learning models also show promise for complex data patterns and larger datasets.

Accuracy Achieved

Reported performance varies by algorithm and dataset, summarized below:

Algorithm	**Typical Accuracy / AUC Range**	**Remarks**
Logistic Regression	70-85%	Simple, interpretable baseline model
Decision Tree	75-83%	Easy to interpret but prone to overfitting
Random Forest	85-97%	Robust ensemble, handles non-linear data well
Gradient Boosted Trees (GBM)	87-95%	Boosting improves accuracy, sensitive tuning
XGBoost	90-99%	Efficient gradient boosting, top performer
SVM	75-90%	Effective with high-dimensional data
Deep Learning (ANN, CNN, RNN)	82-95%+	Captures complex patterns, needs more data

Business Impact and Recommendations

- Accurate churn prediction models allow businesses to target retention efforts efficiently, reducing customer loss and boosting profitability.
- The paper emphasizes using profit-based evaluation metrics alongside traditional accuracy measures to align predictive models with business goals.
- Combining different feature types (demographic, behavioral, social) and adopting explainable AI methods is recommended to improve both model performance and business acceptance.
- There's a research gap in widely accessible, up-to-date datasets, and generalization of models across industries remains challenging.

Research paper 3:

Customer Churn Prediction Using Machine Learning Approaches

[Researchpaper3]

Key Algorithms Used

The study applied and compared performance of the following machine learning algorithms on a telecom churn dataset:

- Decision Tree (DT)
- Random Forest (RF)

Both algorithms were evaluated with and without the use of SMOTE-ENN (Synthetic Minority Over-sampling Technique combined with Edited Nearest Neighbour) to address class imbalance in the churn dataset.

Accuracy Achieved

- Without sampling techniques, Decision Tree achieved an accuracy of 78% and Random Forest 80%, but both performed poorly in predicting the minority "Churn" class due to the imbalanced data.
- With SMOTE-ENN sampling, Decision Tree accuracy improved to 93% and Random Forest to 95%, achieving significantly better recall, precision, and F1-score for the "Churn" class.
- Random Forest combined with SMOTE-ENN produced the best overall prediction with an F1-score of 95%.

Business Impact

- Improved churn prediction models enable telecom companies to proactively identify at-risk customers and apply retention strategies to maximize revenue.
- Addressing the data imbalance problem through SMOTE-ENN is crucial for accurate churn classification, especially for minority churners.
- The results suggest incorporating advanced sampling and ensemble learning methods enhances predictive capabilities crucial for telecom business decisions.

Algorithm	Sampling Technique	Accuracy (%)	Precision (Churn) (%)	Recall (Churn) (%)	F1-Score (Churn) (%)
Decision Tree	None	78	60	54	57
Random Forest	None	80	68	56	58
Decision Tree	SMOTE-ENN	93	93	94	93

Algorithm	Sampling Technique	Accuracy (%)	Precision (Churn) (%)	Recall (Churn) (%)	F1-Score (Churn) (%)
Random Forest	SMOTE-ENN	95	93	97	95

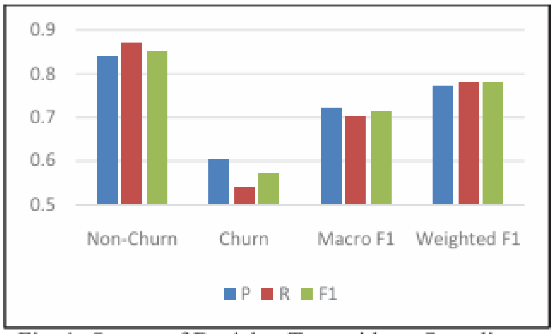

Fig. 1. Output of Decision Tree without Sampling Techniques

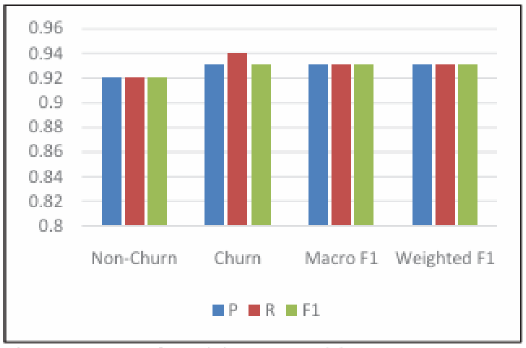

Fig. 3. Output of Decision Tree with SMOTE-ENN

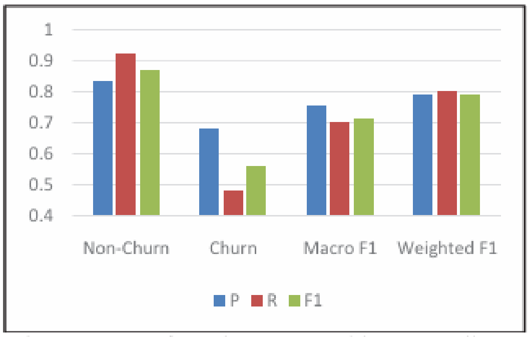

Fig. 2. Output of Random Forest without Sampling Techniques

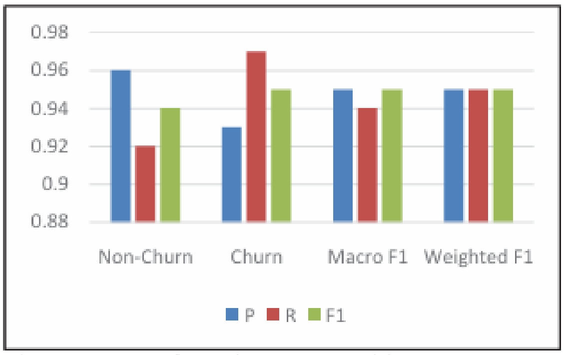

Fig. 4. Output of Random Forest with SMOTE-ENN

3. Dataset Description

- The project uses the **Telco Customer Churn Dataset** available on **Kaggle**.
- It contains information about telecom customers, including their account details, services subscribed, and whether they have churned or not.
- The dataset helps analyse which factors contribute most to customer churn.

Key Details:

- **Source:** Kaggle (Telco Customer Churn Dataset)
- **Total Records:** 7,043
- **Total Features:** 20 (independent variables)
- **Target Variable:** *Churn* (Yes / No)

Main Feature Categories:

- **Demographic Information:** Gender, Senior Citizen, Partner, Dependents
- **Account Information:** Tenure, Contract Type, Payment Method, Paperless Billing
- **Service Details:** Internet Service, Streaming TV, Tech Support, Online Security
- **Financial Information:** Monthly Charges, Total Charges

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
customerI	gender	SeniorCitiz	Partner	Dependen	tenure	PhoneServ	MultipleLi	InternetSe	OnlineSecu	OnlineBac	DevicePro	TechSuppo	Streaming	Streaming	Contract	PaperlessB	PaymentM	MonthlyCh	TotalCharg	Chur
7590-VHV	Female	0	Yes	No	1	No	No phone	DSL	No	Yes	No	No	No	No	Month-to-	Yes	Electronic	29.85	29.85	No
5575-GNV	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed che	56.95	1889.5	No
3668-QPY	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-	Yes	Mailed che	53.85	108.15	Yes
7795-CFO	Male	0	No	No	45	No	No phone	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank trans	42.3	1840.75	No
9237-HQI	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-	Yes	Electronic	70.7	151.65	Yes
9305-CDS	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-	Yes	Electronic	99.65	820.5	Yes
1452-KIO	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No	Month-to-	Yes	Credit car	89.1	1949.4	No
6713-OKO	Female	0	No	No	10	No	No phone	DSL	Yes	No	No	No	No	No	Month-to-	No	Mailed che	29.75	301.9	No
7892-POO	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to-	Yes	Electronic	104.8	3046.05	Yes
6388-TAB	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank trans	56.15	3487.95	No
9763-GRS	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to-	Yes	Mailed che	49.95	587.45	No
7469-LKB	Male	0	No	No	16	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Credit car	18.95	326.8	No
8091-TTV	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	One year	No	Credit car	100.35	5681.1	No
0280-XJGE	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-	Yes	Bank trans	103.7	5036.3	Yes
5129-JLPIS	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes	Yes	Yes	Month-to-	Yes	Electronic	105.5	2686.05	No
3655-SNQ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit car	113.25	7895.15	No
8191-XWS	Female	0	No	No	52	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	One year	No	Mailed che	20.65	1022.95	No
9959-WOF	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank trans	106.7	7382.25	No
4190-MFL	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to-	No	Credit car	55.2	528.35	Yes
4183-MYF	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	Yes	No	No	Yes	Month-to-	Yes	Electronic	90.05	1862.9	No
8779-QRD	Male	1	No	No	1	No	No phone	DSL	No	No	Yes	No	No	Yes	Month-to-	Yes	Electronic	39.65	39.65	Yes
1680-VDC	Male	0	Yes	No	12	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	One year	No	Bank trans	19.8	202.25	No
1066-JKSG	Male	0	No	No	1	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Month-to-	No	Mailed che	20.15	20.15	Yes
3638-WEA	Female	0	Yes	No	58	Yes	Yes	DSL	No	Yes	No	Yes	No	No	Two year	Yes	Credit car	59.9	3505.1	No
6322-HRP	Male	0	Yes	Yes	49	Yes	No	DSL	Yes	Yes	No	Yes	No	No	Month-to-	No	Credit car	59.6	2970.3	No

4.Methodology

❖ Logistic Regression

Overview:
Logistic Regression is a statistical model used for binary classification problems. It estimates the probability that a given input belongs to a particular class (in this case, churn or non-churn) using a logistic function.

Advantages:

- Simple and easy to interpret.
- Performs well when there is a linear relationship between the features and the target variable.

Limitations:

- Limited ability to capture non-linear relationships.
- Performance may drop with highly imbalanced data.

❖ Decision Tree Classifier

Overview:
A Decision Tree is a flowchart-like structure where nodes represent decisions based on feature values, branches represent outcomes, and leaves represent class labels.

Advantages:

- Easy to understand and interpret.
- Handles both categorical and numerical data.

Limitations:

- Prone to overfitting on small or noisy datasets.
- Small data variations can lead to different tree structures.

❖ Random Forest Classifier

Overview:
Random Forest is an ensemble learning algorithm that builds multiple Decision Trees and combines their outputs to improve predictive performance and reduce overfitting.

Advantages:

- Reduces overfitting through averaging.
- Performs well on large and complex datasets.

Limitations:

- Less interpretable compared to single trees.
- Requires more computational resources.

- ❖ **Gradient Boosting Classifier**

Overview:
Gradient Boosting builds models sequentially, where each new model attempts to correct the errors of the previous ones.

Advantages:

- Handles complex relationships effectively.
- Provides high predictive performance.

Limitations:

- Sensitive to hyperparameters and learning rate.
- Longer training time compared to simpler models.

- ❖ **XGBoost Classifier**

Overview:
XGBoost (Extreme Gradient Boosting) is an optimized version of Gradient Boosting that focuses on speed, accuracy, and overfitting prevention through regularization.

Advantages:

- High performance and fast execution.
- Automatically handles missing values and regularization.

Limitations:

- Requires careful parameter tuning.
- Computationally demanding for very large datasets.

- ❖ **Imbalance Handling Technique SMOTE-ENN**

Overview:
A hybrid technique combining SMOTE (synthetic oversampling) and ENN (undersampling by removing noisy samples).

Advantages:
• Balances classes effectively
• Removes ambiguous or noisy points

Limitations:
• Slight increase in training time
• Possible introduction of synthetic noise

➢ Model Explainability (SHAP)

• Purpose of SHAP

SHAP was used to interpret the XGBoost model and understand how each feature contributes to churn predictions, ensuring transparency and trust.

• Global Explainability

Generated through batch predictions in the Streamlit app:

- **Beeswarm Plot:** Displays overall feature influence.
- **Feature Importance Plot:** Ranks features by impact.
- **Dependence Plots:** Shows how key features affect churn probability.
- **Waterfall Plot:** Explains the prediction for the highest-risk customer.

Key Insights: Contract type, tenure, monthly charges, payment method, and technical/security services were major churn drivers.

• Local Explainability

For single-customer inputs, SHAP provides personalized explanations showing which features increased or decreased the likelihood of churn.

➢ Deployment Components

• Streamlit Web Application

The final system was deployed using Streamlit, offering:

- Single and batch prediction
- Automatic SHAP visualizations
- Dashboard for viewing SHAP images
- ROI calculator
- User-friendly interface with tab-based navigation

• ROI Calculator

Based on model predictions, it computes:

- Estimated customers retained
- Revenue saved
- ROI percentage
- Impact on customer lifetime value (CLV)

5.System Design and Architecture

5.1 Overview

The **Customer Churn Prediction System** is designed as a modular and scalable **machine learning (ML) pipeline** that systematically transforms raw customer data into meaningful business insights.
Each phase of the architecture performs a specific function — from data collection and cleaning to model training and interpretability — ensuring reliability, maintainability, and transparency of results.

The architecture comprises the following major components:

1.Dataset Acquisition and Preprocessing:
The system begins with collecting data from the Telco Customer Churn Dataset (Kaggle). The dataset is imported into the Python environment, cleaned to handle missing values and inconsistencies, and encoded for machine learning readiness. This step ensures the dataset is structured, normalized, and suitable for analysis.

2. Exploratory Data Analysis (EDA):
In this phase, data is visualized and statistically analyzed to uncover trends, patterns, and correlations related to churn behavior. Visualizations such as histograms, box plots, and heatmaps are generated to understand relationships between numerical and categorical variables, detect outliers, and study churn distribution across different customer segments.

3.Feature Engineering:
New variables are derived from existing data to enhance model performance. Examples include creating interaction features (e.g., tenure × monthly charges) and converting categorical variables into numeric formats using label or one-hot encoding. Highly correlated or low-variance features are removed to reduce noise and overfitting.

4.Model Training and Evaluation:
Several supervised learning algorithms — Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost — are trained on the processed data. Model performance is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
The goal is to select the most balanced and interpretable model for predicting customer churn.

5. Hyperparameter Tuning:
After model training, hyperparameter tuning is performed to optimize the performance of each algorithm. This process involves systematically adjusting key parameters such as the number of estimators, learning rate, maximum depth, and regularization terms to achieve the best balance between accuracy, recall, and generalization. Randomized Search was used to identify the optimal combination of parameters for models such as Random Forest and XGBoost. This stage ensures that the final models deliver improved predictive performance without overfitting the data.

6. Model Explainability (SHAP Integration):
SHAP was integrated to interpret the final XGBoost model by identifying how each feature contributes to churn predictions. It provides both global and local explanations, offering insights into overall feature importance as well as individual customer predictions. In the Streamlit application, SHAP generates beeswarm plots, feature importance charts, dependence plots, and waterfall plots to visually explain model decisions. This helps highlight key churn drivers such as contract type, tenure, monthly charges, and payment method.

5.2 Workflow

The overall workflow of the project is as follows:

1. Dataset Acquisition
- Collect the *Telco Customer Churn* dataset from Kaggle.
- Import it into the Python environment for analysis.

2. Data Preprocessing
- Handle missing and duplicate values.
- Encode categorical variables and scale numerical features.
- Balance the dataset using SMOTE-ENN to mitigate class imbalance and enhance model performance on the churn class.

3. Exploratory Data Analysis (EDA)
- Analyze customer patterns and churn distribution.
- Visualize relationships between features such as tenure, monthly charges, and contract type.
- Identify key trends and correlations affecting churn.

4. Feature Engineering
- Create new meaningful features to improve model accuracy..
- Prepare the final feature set for model training.

5. Model Training
- Train multiple machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost.
- Tune hyperparameters and optimize model performance.

6. Model Evaluation
- Evaluate models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- Compare results to select the most reliable model for churn prediction.

7. Model Explainability
- Identify the most important features influencing churn using feature importance and SHAP analysis.
- Interpret model behavior to provide transparent and business-relevant insights.

8. Dashboard and ROI Calculator
- Create an interactive dashboard in Power BI or Tableau to visualize churn insights.
- Develop an ROI calculator in Streamlit to estimate the financial impact of reducing churn.

9. Model Deployment
- Deploy the final trained model using Streamlit .
- Integrate it with the dashboard and ROI tool to form a complete decision-support system.

5.3 Tools and Technologies

Category	Tools / Technologies Used	Purpose / Description
Programming Language	Python (Version 3.10 or above)	Python is used as the core programming language due to its simplicity, readability, and large support for machine learning and data analysis libraries.
Data Manipulation Libraries	Pandas, NumPy	Used for importing, cleaning, and processing the dataset. These libraries help in handling large data tables efficiently and performing numerical computations.
Visualization Tools	Matplotlib, Seaborn	Used during Exploratory Data Analysis (EDA) to create charts, correlation heatmaps, and visual summaries that highlight churn trends and patterns.
Machine Learning Libraries	scikit-learn, XGBoost	Provide algorithms for training and evaluating models such as Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Also used for performance measurement and model comparison.
Model Explainability Tools	SHAP	Used to interpret the XGBoost model with global and local explanations, generating beeswarm, bar, dependence, and waterfall plots.
Deployment Framework	Streamlit	Used to deploy the final system with single & batch predictions, SHAP visualization dashboard, and ROI calculator.
Development Environment	Jupyter Notebook, VS Code	Interactive platforms used for coding, running experiments, visualizing results, and documenting the workflow.
Version Control (Planned)	Git / GitHub	To maintain version history of code, enable collaboration among team members, and store the final implementation securely.

6. RESULTS AND DISCUSSION

The models were trained and evaluated on the processed Telco Customer Churn dataset using various performance metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The test set performance of all five models is summarized below:

6.1 Model Comparison Summary (Test Set)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.6984	0.4634	0.8636	0.7160	0.8347
Decision Tree	0.7126	0.4746	0.7754	0.7286	0.7534
Random Forest	0.7310	0.4959	0.7995	0.7458	0.8318
Gradient Boosting	0.7324	0.4975	0.7888	0.7469	0.8337
XGBoost	**0.7367**	**0.5026**	0.7754	**0.7505**	0.8278

6.2 Best Models by Metric

- **Best Accuracy:** XGBoost (0.7367)
- **Best Precision:** XGBoost (0.5026)
- **Best Recall:** Logistic Regression (0.8636)
- **Best F1-Score:** XGBoost (0.7505)
- **Best ROC-AUC:** Logistic Regression (0.8347)

6.3 Model Performance Analysis

Logistic Regression achieved the highest recall (0.8636) and ROC-AUC (0.8347), indicating strong sensitivity in identifying churners. However, it suffered from the lowest precision (0.4634), leading to a higher false positive rate.

To address the inherent class imbalance in the churn dataset, SMOTE-ENN was applied during preprocessing, enabling all models to achieve more balanced precision-recall trade-offs and improving recall on the minority class.

Random Forest and Gradient Boosting delivered balanced performance with competitive recall, precision, and AUC scores, demonstrating robustness and generalization.

XGBoost emerged as the top performer in accuracy (0.7367), precision (0.5026), and F1-score (0.7505), while maintaining strong recall (0.7754). Its F1-score superiority indicates the best balance between precision and recall—a critical requirement for churn prediction where both false positives and false negatives carry business costs.

6.4 Justification for Selecting XGBoost

Although Logistic Regression excelled in recall, its low precision would result in inefficient retention campaigns targeting many non-churners. In contrast, XGBoost provides the optimal trade-off:

1. **Highest Accuracy & F1-Score** → Better overall predictive reliability.
2. **Best Precision** → Reduces wasted resources on false positives.
3. **Strong Recall (0.7754)** → Still captures the majority of true churners.
4. **Business Alignment** → Maximizes ROI by focusing retention efforts on **high-probability churners** with fewer misidentifications.
5. **SHAP Compatibility** → Seamlessly integrates with explainability tools for actionable insights.

Thus, XGBoost was selected as the final model for deployment, balancing predictive performance, interpretability, and business impact.

7.Conclusion

The customer churn prediction system was successfully developed using the Telco Customer Churn dataset and implemented through a complete end-to-end machine learning workflow. Multiple models—including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost—were trained and evaluated to identify the most suitable algorithm for predicting churn.

Among these, **XGBoost emerged as the best-performing model**, achieving the highest test accuracy (**73.67%**), precision (**0.5026**), and F1-score (**0.7505**), while maintaining strong recall (**0.7754**). Although Logistic Regression yielded the highest recall (**0.8636**), its lower precision made it less suitable for cost-effective retention campaigns. XGBoost provided the optimal balance between identifying true churners and minimizing false positives—a critical requirement for business deployment.

To enhance interpretability, the system integrates **SHAP explainability**, providing both global and local insights into model predictions. SHAP visualizations such as beeswarm plots, feature importance charts, dependence plots, and waterfall explanations helped identify key churn drivers including **contract type, tenure, monthly charges, and payment method**.

The final solution was deployed through an interactive **Streamlit application**, which supports single and batch predictions, automatic SHAP visualization, and a dedicated dashboard for analysis. Additionally, an **ROI Calculator** was implemented to estimate the financial benefits of retention strategies by calculating potential customers saved, revenue retained, and CLV impact.

Overall, the project demonstrates that ensemble models combined with explainability and deployment tools can provide accurate, interpretable, and business-oriented solutions for customer churn prediction. The developed system not only identifies high-risk customers but also supports real-world decision-making through transparent insights and ROI-driven retention planning.

8.Future Work

While the current project successfully delivers an end-to-end churn prediction system — including model training, SHAP explainability, Streamlit deployment, batch processing, and ROI calculation — there are several opportunities for further enhancement and expansion.

- **Cloud Deployment:** Host the Streamlit app on AWS/GCP/Azure for real-time access and scalability.
- **CRM Integration:** Connect the model with CRM systems to trigger automated churn alerts and retention actions.
- **Advanced Segmentation:** Add customer segmentation (e.g., RFM or clustering) to target specific groups more effectively.
- **Enhanced CLV Modeling:** Improve the ROI module by integrating more accurate and predictive CLV calculations.
- **Power BI/Tableau Dashboards:** Build richer dashboards for visualizing churn trends and financial impact.
- **Automated Retraining:** Implement a pipeline that retrains the model as new customer data becomes available.

9.Reference

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